BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 212 · Timed Paper 2 algorithms practice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which structure retrieves the most recently added item first?

Q2. What is the sorted-input precondition?

Q3. What moves right as a key is inserted into a sorted prefix?

Q4. What are the worst-case time complexities of linear and binary searches?

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I Do · Read new knowledge

Key vocabulary

trace — Record an algorithm's changing state.

sorted prefix — Part already in order during insertion sort.

precondition — Requirement, such as ordering, for an algorithm.

O(log n) — Work grows by repeated halving.

auxiliary space — Extra storage required by an algorithm.

New learning

Read the task command — A complete pass means all permitted adjacent comparisons for that pass, not just one swap.

Trace one pass — [6,2,5,1] becomes [2,5,1,6] after comparing each pair.

Finish every pass — Continue to [2,1,5,6], then [1,2,5,6]. A clear trace shows each pass.

Worked example

Model the first complete ascending bubble-sort pass on [4,1,3].

1. Predict first — Start [4,1,3] and compare 4 with 1: swap.

2. Check the method — Now compare 4 with 3: swap, producing [1,3,4].

3. Check the method — The pass is complete and the array is sorted.

We Do · Practise with guidance

Trace one full bubble-sort pass on [7,2,6,3].

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Record the array after each adjacent comparison that causes a swap.
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Predict first

7 swaps with 2 giving [2,7,6,3]; then 7 swaps with 6 giving [2,6,7,3].

Check the method

Compare 7 with 3: swap to [2,6,3,7].

The complete first pass ends [2,6,3,7], not yet sorted.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

An array contains [8,3,6,1]. Trace ascending insertion sort, giving the array after each outer-loop iteration and explaining how the key is positioned.

Question 2

Write a C# function BinarySearch(int[] a,int target) returning the first matching index it encounters, or −1. Include correct bounds and a sorted-input precondition.

Question 3

Explain one advantage and one disadvantage of merge sort over bubble sort, including time complexity, and identify an input scenario for which linear search is more practical than binary search.

Exit Ticket · Answer before checking

Binary search works correctly on any unordered input.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Stack (LIFO).

Q2. The array must be sorted.

Q3. All preceding values greater than the key.

Q4. Linear O(n); binary O(log n) for sorted input.

You Do pages · Independent questions

Q1. [8] After inserting key 3: [3,8,6,1] [2]; key 6: [3,6,8,1] [2]; key 1: [1,3,6,8] [2]; each key is inserted into the sorted prefix by shifting larger values right [2].

Q2. [8] int low=0,high=a.Length-1 [2]; while(low<=high) [1]; mid=low+(high-low)/2 [1]; if(a[mid]==target)return mid [1]; if(a[mid]<target)low=mid+1; else high=mid-1 [2]; return -1 [1]. Input must be sorted.

Q3. [8] Merge sort O(n log n) worst case [2] versus bubble $O(n^2)$ [2]; merge sort typically needs O(n) extra array space [2]; linear search is suitable for a small unsorted collection where sorting before a single search would cost time [2].

Exit Ticket · Model answer

False. It requires a sorted search sequence.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding